

Cambridge IGCSE™

PHYSICS**0625/42**

Paper 4 Theory (Extended)

May/June 2024

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **12** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Acronyms and shorthand in the mark scheme

Acronym / shorthand	Explanation
A mark	Final answer mark which is awarded for fully correct final answers including the unit.
C mark	Compensatory mark which may be scored when the final answer (A) mark for a question has not been awarded.
B mark	Independent mark which does not depend on any other mark.
M mark	Method mark which must be scored before any subsequent final answer (A) mark can be scored.
Brackets ()	Words not explicitly needed in an answer, however if a contradictory word / phrase / unit to that in the brackets is seen the mark is not awarded.
<u>Underlining</u>	The underlined word (or a synonym) must be present for the mark to be scored. If the word is a technical scientific term, the word must be there.
/ or OR	Alternative answers any one of which gains the credit for that mark.
owtte	Or words to that effect.
ignore	Indicates either an incorrect or irrelevant point which may be disregarded, i.e., <u>not</u> treated as contradictory.
insufficient	An answer not worthy of credit <u>on its own</u> .
CON	An incorrect point which contradicts any correct point and means the mark cannot be scored.
ecf [question part]	Indicates that a candidate using an erroneous value from the stated question part must be given credit here if the erroneous value is used correctly here.
cao	Correct answer only.
ORA	Or reverse argument.

Question	Answer			Marks
1(a)	0.077 kg OR 77 g			A2
	$g = W / m$ OR ($m =$) W / g OR 0.75 / 9.8			C1
1(b)	2 <u>vectors</u> at right angles	OR	use of Pythagoras' theorem e.g. $a^2 + b^2 = c^2$ OR (force =) $\sqrt{1.2^2 + 0.75^2}$	B1
	1.4 (N)			B1
	58(°)			A2
	resultant force including correct direction of arrow	OR	use of trigonometry to find angle e.g. $\tan \theta = 1.2 / 0.75$	C1
1(c)	any two from: • velocity • speed • direction • acceleration / deceleration • moment			B2

Question	Answer	Marks
2(a)	change in velocity per unit time OR rate of change of velocity OR ($a =$) $\Delta v / \Delta t$	B1
2(b)(i)	12 s	A2
	($\Delta t =$) $\Delta v / a$ OR 13 / 1.1	C1
2(b)(ii)	570 000 N	A2
	$F = ma$ OR ($F =$) ma OR ($F =$) $520\,000 \times 1.1$	C1
2(b)(iii)	(additional force is needed to overcome) friction OR air resistance OR drag	B1

Question	Answer	Marks
3(a)	<u>gravitational</u> (potential) energy (store before / as the ball falls)	B1
	kinetic energy (store) <u>increases</u> (as the ball falls) OR energy transferred to kinetic energy (store as the ball falls)	B1
	(energy transferred from) kinetic energy (store) to internal / thermal energy (store)	B1
3(b)	5.9 m / s	A3
	(Δ) $E_p = (\Delta)E_k$ OR E_p lost = E_k gained OR gravitational potential energy lost = kinetic energy gained OR $mg(\Delta)h = \frac{1}{2}mv^2$	C1
	$v^2 = 2g(\Delta)h$ OR $v^2 = 2 \times 9.8 \times 1.8$ OR $v^2 = 35(.28)$	C1

Question	Answer	Marks
4(a)	(evaporation:) (only) at the surface OR boiling: happens throughout the liquid	B1
	(evaporation:) takes place at any temperature OR boiling: takes place at a specific temperature / boiling point	B1
4(b)(i)	113 (K)	B1
4(b)(ii)	conduction	B1
	convection	B1
4(c)	particles collide with the walls / container	B1
	(particles) exert a force on the walls OR collision with walls produces a change in momentum (of particles)	B1
	pressure is force per unit area OR $p = F / A$ OR pressure is rate of change of momentum per unit area	B1

Question	Answer	Marks
5(a)	ray travels along the normal OR angle of incidence = 0(°)	B1
5(b)(i)	$n = 1 / \sin c$ OR $(n =) 1 / \sin c$ OR $(n =) 1 / \sin 42(^{\circ})$	M1
	1.5	A1
5(b)(ii)	ray reflected at BC AND no refracted ray	M1
	ray hits AC with angle of incidence = 0°	A1
	correct refraction of candidate's ray into air at AC	B1

Question	Answer	Marks
5(c)	any two from: • high rates (of data transmission) / fast (data transmission) • carry large amounts (of data / information) • secure • little signal / data loss • glass is transparent to (some) infrared	B2

Question	Answer	Marks
6(a)(i)	C marked and labelled at a peak of the sine wave	B1
	R marked and labelled at a trough of the sine wave	B1
6(a)(ii)	graph / it does not show (variation with) displacement	B1
6(a)(iii)	(amplitude) increases AND (frequency) decreases	B1
6(b)	0.12 m	A3
	$v = f \lambda$ OR $(\lambda =) v / f$ OR $(\lambda) = 1500 / 13\,000$	C1
	1500 / 13 000 OR 1.2×10^N	C1
6(c)	330 m / s $\leq$ speed $\leq$ 350 m / s	B1

Question	Answer	Marks
7(a)	(end of) one piece of steel brought close to (the end of) another piece of wire	B1
	look to see if there is repulsion/attraction AND test between different ends/poles of wire	B1
	any two from: - repeat a valid test between the other pieces - only magnets repel each other OR the pieces that repel are magnets - attractions at both ends indicates one of them is unmagnetised OR the piece that only attracts is unmagnetised OR the piece that does not repel (at both ends) is unmagnetised	B2
7(b)(i)	0.19 A	A3
	$I_p V_p = I_s V_s$ OR $(I_p =) I_s V_s / V_p$ OR $(I_p =) 45 / 240$ OR $I_p V_p = 45$ OR power in primary = power in secondary	C1
	$(I_p =) 45 / 240$	C1
7(b)(ii)	labelled diagram showing: (soft) <u>iron</u> core <u>copper</u> (coils) - primary and secondary (coils) labelled AND fewer coils on secondary than on primary	B3

Question	Answer	Marks
8(a)(i)	correct voltmeter symbol connected across LDR	B1
8(a)(ii)1	(resistance) increases	B1
8(a)(ii)2	(p.d.) increases because resistance of parallel combination of LDR and LED increases	B1
	greater proportion of (total) p.d. across LDR / LED / parallel combination of LDR and LED	B1

Question	Answer	Marks
8(b)	960 Ω	A3
	current in each bulb = 0.25 OR $R = V / I$ OR $(R =) V / I$	C1
	resistance = $240 / 0.25$ OR $1 / 480 = 1 / R + 1 / R$	C1

Question	Answer	Marks
9(a)(i)	any two from: • reduce exposure time AND low(er) amount of radiation absorbed • increase distance between source and hospital staff AND lower amount of radiation reaches staff • use of shielding (e.g. walls, lead etc.) AND radiation absorbed by shielding / cannot penetrate through shielding • use isotopes with short half-life AND lower amount of radiation emitted from patient / radiation (above background) emitted for a shorter period of time • use of film badge / dosimeter AND manage individuals exposure owtte • restrict pregnant staff / patient in hospital AND radiation may harm foetus owtte	B2
9(a)(ii)	high ionisation (within body)	B1
	radiation would not reach detector (outside body)	B1
9(b)	${}^{24}_{11}\text{Na} \rightarrow {}^{24}_{12}\text{Mg} + {}^0_{-1}\beta$	
	${}^{24}_{11}\text{Na}$ on LHS	B1
	${}^0_{-1}\beta$ on RHS	B1
	${}^{24}_{12}\text{Mg}$ on RHS	B1

Question	Answer	Marks
10(a)	any two from: • minor planets OR dwarf planets • comets • asteroids	B2
10(b)	elliptical	B1
10(c)	kinetic energy (store) decreases AND potential energy (store) increases (as object moves from A to B)	B1
	energy is conserved	B1
10(d)	$2.6 \times 10^3 \text{ s}$	A2
	$v = s / t$ OR $(t =) s / v$ OR $7.8 \times 10^{11} / 3.0 \times 10^8$	C1
11(a)	Milky Way	B1
11(b)	Big Bang (Theory)	B1
11(c)(i)	shortly after the Universe was formed	B1
11(c)(ii)	Universe has expanded	B1
	(radiation) has been redshifted (to the microwave region of the electromagnetic spectrum)	B1